

MUTHUKKUMAR.A.L

Mechanical BIM Modeler

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Profile

Motivated Mechanical BIM Modeler with 11 months of hands-on experience in HVAC systems and BIM coordination. At Pinnacle Infotech, I played a key role in international projects, including luxury hotels and commercial buildings in Saudi Arabia, leveraging Revit, AutoCAD, and Navisworks to produce accurate models, detailed shop drawings, and resolve critical clashes. Known for simplifying complex design challenges, enhancing coordination workflows, and consistently delivering high-quality results within tight deadlines. Dedicated to continuous learning and thriving in collaborative, fast-paced project environments.

Professional Experience

PINNACLE INFOTECH,

07/2024 – present | MADURAI, INDIA

GRADUATE ENGINEERING TRAINEE

HVAC Engineer skilled in BIM design using Revit, AutoCAD, and Navisworks. Experienced in modeling complex systems, resolving clashes, and reducing errors. Strong collaborator who improves coordination and consistently delivers projects on time.

- Spearheaded the complete 3D HVAC modeling of 3 floors for the Rose Hotel project (Jeddah, Saudi Arabia) using Revit by independently interpreting client-provided IFC files, accelerating delivery through proactive clash resolution with MEP services in Navisworks.
- Produced detailed HVAC shop drawings by utilizing schematic inputs and BIM coordination data, enhancing clarity and accuracy to reduce on-site installation issues.
- Resolved critical clashes using Navisworks through comprehensive multi-disciplinary coordination reviews, minimizing rework and ensuring smooth integration between HVAC systems and other building services.
- Developed HVAC schematics in AutoCAD by translating design intent into clear, serviceable diagrams, improving team understanding and streamlining the design approval process.
- Orchestrated seamless coordination with cross-functional BIM teams to resolve interdisciplinary conflicts, boosting project alignment and reducing coordination meeting time.
- Scrutinized HVAC models against project standards and QA checklists, elevating model accuracy and minimizing client revision requests.
- Mastered Revit, AutoCAD, and Navisworks to execute end-to-end HVAC BIM tasks, cutting modeling and documentation time through tool-specific automation and shortcuts.
- Contributed to over 2 international HVAC projects including luxury hotels and commercial complexes across Saudi Arabia and the UAE, consistently meeting client deadlines with on-time submissions.

Education

B.E MECHANICAL ENGINEERING - 77%,
GOVERNMENT COLLEGE OF ENGINEERING

07/2020 – 05/2024 | ERODE, INDIA

HSC - 61%,
ROSEMARY MATRIC HR.SEC.SCHOOL,RAMNAGAR

TIRUNELVELI, INDIA

SSLC - 81%,
ROSEMARY MATRIC HR.SEC.SCHOOL,RAMNAGAR

TIRUNELVELI, INDIA

Skills

Technical Skills:

- HVAC Design & Modeling (Revit, AutoCAD)
- Clash Detection & Resolution (Navisworks)
- BIM Coordination & Collaboration
- Shop Drawing Preparation
- IFC Interpretation & Schematic Drafting
- Quality Assurance & Compliance Checks

Soft Skills:

- Problem Solving & Analytical Thinking
- Team Collaboration & Communication
- Time Management & Deadline Orientation
- Mentoring & Knowledge Sharing

Software Proficiency

- AutoCAD
- Revit
- Navisworks
- McQuay Duct Sizer

Declaration

I hereby declare that all the information given above is true and correct to the best of my knowledge

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